



Mitesh Upadhyay Lead Data Scientist



Professional Summary:

- Innovative and results-driven AI Architect with over 13 years of experience in designing and delivering solutions.
- Expert in Generative AI and Agentic AI solutions across diverse industries. Adept at architecting intelligent ecosystems that combine, RAG (Retrieval-Augmented Generation), autonomous multi agent frameworks, MCP and A2A(Agent to Agent) protocol to knowledge orchestration, and decision intelligence.
- Proven expertise in leveraging LLM orchestration frameworks, prompt engineering, and AI agent collaboration to build high-performing systems. Skilled in integrating cloud-native AI stacks (Azure, AWS, GCP) with vector databases, embeddings for adaptive reasoning and contextual intelligence, along with leveraging AWS Bedrock, Azure AI Foundry, Google Vertex AI, and CrewAI frameworks.
- Expertise in MLOps, LLMOps, DevOps, Inference and Training Data Pipelines Management/Ingestion, Automation of data movement, transformation, quality and integrity verification , production drift monitoring and developing end to end CI/CD pipelines, developing & deploying scalable models, orchestration, scheduling, automation, ML Operations.
- Conduct exploratory data analysis (EDA) to uncover trends, patterns, and relationships in data, overseeing data acquisition, analysis, and generating valuable insights.
- Thorough understanding of prompt engineering; implementation exposure to LLM agents like Lang Chain & vector databases, RAG implementation .
- Knowledge of the latest techniques in this space such as LORA adaptors, PEFT, quantization, fine tuning LLMs including BERT and GPT based.
- Expertise in generative AI models (like GPT4, Claude 3, Turbo3.5 , DeepSeek, Llama etc). Also exposure to integrating LLMs(Open AI GPT) using API endpoints.

Skills Snapshot:

Technical Skills	• Generative AI, Agentic AI, LLMs, RAG , Multi-Agent Frameworks, Prompt Engineering, Fine-Tuning GPT-4, Claude 3, DeepSeek, LLaMA, Turbo 3.5, LangChain, CrewAI, MCP , LLM Orchestration, Multi-Agent Collaboration, Vector Databases, Embeddings, Semantic Search, • Azure AI Foundry, Azure AI Vision, Azure AI Speech, Azure Document Intelligence, Azure Container Apps, Azure AI Search, Microsoft Copilot Studio , Azure Machine Learning, AWS Bedrock, AWS, MLOps, LLMOps, CI/CD Pipelines, Model Deployment.
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	• MLOps, LLMOps, CI/CD Pipelines, Model Deployment, Production Drift Monitoring, Inference Pipelines, Training Data Pipeline Management, Model Orchestration & Scheduling, Automation • Production Drift Monitoring, Inference Pipelines, Training Data Pipeline Management, Model Orchestration, Exploratory Data Analysis (EDA), Data Acquisition & Ingestion, Data Transformation, Data Quality & Integrity Verification, Data Movement Automation, Pattern Recognition & Insight Generation
Functional Skills	• Communication and collaboration • Problem-solving and decision-making • Project management • Strategic thinking • Leadership and stakeholder engagement

Education Details:

Qualification	Educational Institution
Bachelors	BSc (Computer Science), DAVV , Indore
Masters	MCA, RGPV, Indore

Certifications:

AWS Certified Machine Learning Speciality
AWS Certified Solution Architect Associate
AWS Certified Developer Associate
Terraform Certified Associate
PG Diploma in AI/ML from IIITB
Azure AI Fundamental

Detailed Professional Experience:

Current Organization	• Yash Technologies
Previous Organization	• Coforge • John Deere • Tech Mahindra



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Detailed Project Experience:

Project 1	
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Client Name	
Project Name	MutliAgent Quantitative Analysis System
Team Size	6
Role	Sr. Tech lead
Environment	VS CODE, GIT, Azure AI Services ,Azure Container Instances ,Azure DevOps, Azure PostgreSQL, Azure Blob Storage, Azure Managed Identity,Azure Monitor, Azure Application Insights, Docker

Brief Description of Project:

This is a production-grade, full-stack AI platform where multiple autonomous agents collaborate in a pipeline to collect financial data, derive analytical insights, and produce professional investment reports — all without requiring manual intervention between stages.

Core Components

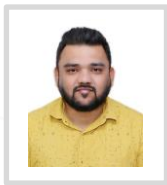
Frontend serves as the user-facing dashboard. It provides controls to trigger analysis runs, select target assets or sectors, monitor the live status of each agent, and download or view finalized reports. The interface is designed for non-technical stakeholders such as analysts and portfolio managers.

FastAPI Backend (Orchestrator) is the central nervous system of the system. It exposes REST endpoints for triggering analysis jobs, manages the lifecycle of each agent, handles authentication and request validation, and coordinates message passing between agents. It is built asynchronously to handle multiple concurrent analysis runs without blocking.

Multi-Agent Layer contains three specialized autonomous agents:

The **Data Scraper Agent** is responsible for acquiring raw market data from external APIs (such as Yahoo Finance or Alpha Vantage), news feeds, SEC filings, and macroeconomic indicators. It normalizes and validates data before passing it downstream.

The **Trend Analyzer Agent** consumes the scraped data and applies quantitative models — including moving averages, volatility metrics, momentum signals, and correlation analysis — to identify patterns and generate structured analytical output.



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The **Report Generator Agent** takes the analytical findings and constructs a professional investment report. It formats content into narrative sections (executive summary, risk assessment, technical outlook, recommendations) and outputs PDF or HTML files.

The agents communicate via a shared context store and message-passing mechanism managed by the orchestrator. Each agent is independently retryable and observable.

Data Persistence

Azure PostgreSQL logs all transaction metadata — every agent run, job status, timestamps, input parameters, and error records. This forms the full audit trail and supports monitoring dashboards and job replay.

Azure Blob Storage acts as the report archive. Every generated report is stored with versioning, making it possible to retrieve historical analyses, compare reports over time, and serve them to the frontend on demand.

Contributions

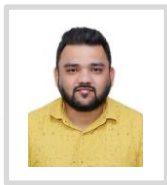
- Designed a multi-agent orchestration framework
- Implemented agent observability and lifecycle management
- Drove containerization and CI/CD pipeline design

Project 2	
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Client Name	Club Mahindra
Project Name	MASO - MutliAgentic Service Orchestrator
Team Size	5
Role	Sr. Tech lead
Environment	VS CODE, GIT, AWS, Docker, AWS Bedrock, Bedrock Knowledgebase, OpenSearch

Brief Description of Project:

Built a Multi-agent AI system that autonomously interprets customer intents, validates policy constraints, and orchestrates backend actions across multiple systems — improving response accuracy and reducing failed bookings.



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Supervisor Agent (Router): Entry point that classifies intent and delegates to specialized collaborator agents.

Collaborator Agents:

Enquiry Agent – Handles general queries, retrieves hotel & policy details via Bedrock Knowledge Base (RAG-based, OpenSearch embeddings over S3 documents).

Booking Agent – Validates credit-day balance, checks room availability, and performs soft bookings via DynamoDB integration.

Support Agent – Manages customer complaints, logs CRM tickets, and routes human callback requests.

Each agent uses dedicated Action Groups (Bedrock Tools) to execute domain-specific tasks — such as querying DynamoDB, fetching from OpenSearch, or posting to the ticketing system. The Bedrock Knowledge Base provides context-grounded responses for policies and FAQs, ensuring factual and compliant interactions.

key Features & Flow:

- Intent classification and routing through Supervisor Agent.
- Policy-aware vacation recommendations and plan validation.
- Real-time balance check and soft booking with a 6-hour payment window.
- Automated complaint logging with ETA tracking and human escalation.
- Contextual retrieval from RAG-powered Bedrock Knowledge Base with OpenSearch embeddings.
- All interactions governed by Bedrock Guardrails (PII redaction, policy compliance).

Contributions:

- Architected the Unified Multi-Agent System
- Implemented the End to End Agentic workflow .
- Cost calculation and optimization.

Project 3	
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Client Name	CIPLA
Project Name	MIRA - Medical Intelligence & Reasoning Agent - (CIPLA Ltd)
Team Size	5



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Role	Senior Tech lead
Environment	VS CODE, GIT, GCP, gTTs, RAG, Gemini

Brief Description of Project:

Designed and developed an advanced Generative AI-powered Voice Chatbot for CIPLA, a leading pharmaceutical company, to enable seamless multilingual interactions with healthcare professionals and patients. The system was fine-tuned on domain-specific medical data, empowering it to understand and respond accurately to complex clinical and pharmaceutical queries through both voice and text modalities.

key Features & Highlights:

Multimodal Architecture:

Integrated image, text, and voice understanding using Vision-Language Models (VLMs) to interpret product labels, prescriptions, and package images along with text-based queries. Models leveraged included Google's Gemini Pro (for multimodal reasoning) and AWS Bedrock Titan Embeddings for semantic search and context retrieval.

Voice-Enabled Conversational Layer:

Implemented Google Speech-to-Text API for multilingual voice input and Google Text-to-Speech for natural voice responses. Supported 10+ Indian and global languages with context-aware tone and pronunciation adaptation.

RAG (Retrieval-Augmented Generation) Framework:

Built a RAG pipeline to fetch up-to-date drug information, dosage guidelines, and regulatory documents. Vectorization: AWS Bedrock embeddings (Titan G1) Storage & Retrieval: AWS OpenSearch Orchestration: LangChain and FastAPI-based backend with context caching

Multilingual NLP Layer:

Utilized Google Translation API for translation and transliteration support across multiple Indian languages (Hindi, Marathi, Tamil, Bengali, etc.) ensuring inclusivity and accessibility.

Contributions:

- Architected the solution
- Engineered a robust RAG pipeline.



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- Orchestrated scalable workflows

Project 4	
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Client Name	
Project Name	Multi-Agent Adaptive Learning & Assessment Platform
Team Size	6
Role	AI Engineer
Environment	VS CODE, AWS, MCP, A2A, Langchain,

Brief Description of Project::

Architected LearnSentra, a multi-agent education intelligence layer for the BYJU'S app that delivers personalized tutoring, adaptive practice, and mastery tracking across K-12 (CBSE/ICSE/State boards) and JEE/NEET prep.

Tutor Agent – Conversational explanations aligned to BYJU'S videos and pedagogy; multilingual (English + Hindi + regional languages)

Assessment Agent – Adaptive quizzes, past-year question drills, concept mastery scoring, item-response theory (IRT) calibration.

Content Curation Agent – Maps syllabus → concepts → BYJU'S video capsules → practice sets; ensures grade/board alignment.

Progress & Recommendation Agent – Detects weak concepts, sequences next best activity (video → practice → revision).

The system uses MCP (Model Context Protocol) to safely invoke BYJU'S internal tools/content APIs and A2A (Agent-to-Agent) collaboration

(LangGraph/CrewAI) to coordinate tutoring, assessment, and recommendations in real time—online and low-bandwidth/offline-first.